

PRIMARY EMOTIONS

Psychologists think there are six primary emotions, each of which causes a characteristic facial expression. These expressions are the same in all the world's people—a smile means the same thing whether you live in the Sahara desert or the Amazon rain forest. Secondary emotions are more complex and include guilt, shame, pride, and envy.



Fear



Disgust



Surprise



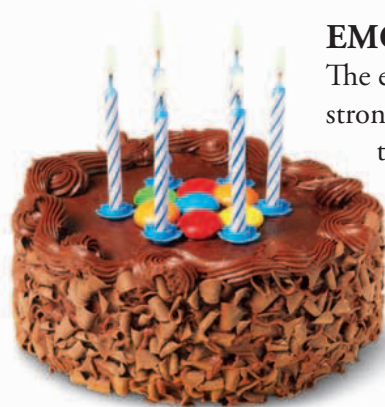
Joy



Anger



Sadness



EMOTIONAL MEMORY

The emotional parts of the brain have strong links with the parts of the brain that store memories. Events that trigger powerful emotions such as fear or joy also form especially vivid, photographic memories. That's why you can probably remember far more details about your last birthday than you can about the day afterward.

PHOBIAS

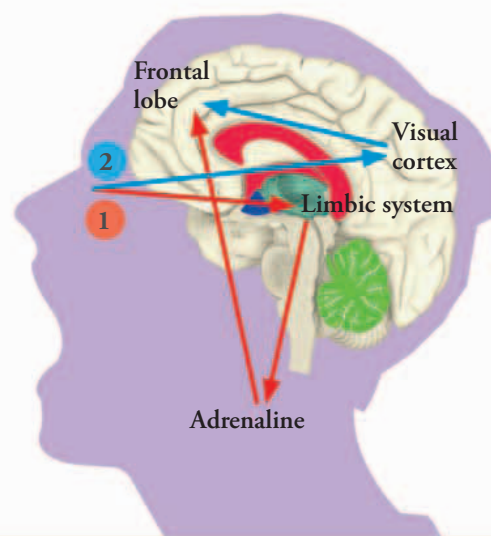
Some people experience a feeling of terror in reaction to something harmless, such as the sight of a spider. Before they have time to think, their brain has put the body on red alert, causing a racing heart, queasy stomach, and feeling of dread. This kind of overreaction is called a phobia. Most phobias are caused by animals, but other triggers include heights, flying, and the sight of blood.



HOW FEAR WORKS

Fear is the most powerful emotion of all. Lightning reactions are vital in times of danger, so the fear signal takes a shortcut from your brain to your body.

- 1 Signals from your eyes pass through part of the brain called the limbic system. This makes a quick visual analysis, and if it spots anything that looks scary, it puts the body on red alert. The hormone adrenaline is released, making you feel fear.
- 2 A slower signal from your eyes reaches the visual cortex at the back of the brain. This makes a detailed analysis of what you've seen. The front of your brain then uses thought and memory to decide whether the threat really is dangerous.



LIE DETECTORS

Your emotions tend to trigger certain physical reactions if you're lying. Your heart speeds up, and the skin on your hand becomes slightly damp with sweat. Lie detector machines monitor these physical reactions, but good liars can stay calm and fool the system.

